LESHEN ZHANG

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EDUCATION

Fudan University 09/2023 – 07/2027 (expected)
Bachelor's Degree in Chemistry | GPA: 3.89/4.00 *Shanghai, China*
Advisor: Prof. Gengfeng Zheng

PUBLICATIONS

‡EQUAL CONTRIBUTION, *CORRESPONDING AUTHOR

[J.1] **Leshen Zhang**, Guangsheng Liu, Xiao Ma, Abdulrahman Allangawi, Wan-Lu Li*. **Defect-Mediated Regulation of Interfacial Hydrophobic Transport via Cavitation Thermodynamics**.
Chem. Sci. **2026**, 17, 15735–15744. [PDF] [DOI] (Sole First Author)

[J.2] Guangsheng Liu‡, Xiao Ma‡, **Leshen Zhang**‡, Jenedith Pascasio, Jonathan Yang, Yuxiang Chen, Wan-Lu Li*. **CatGo: Bridging CLI Coding Agents with Interactive Structure and Workflow Management for Computational Chemistry**.
Digit. Discov. **2026**, 5, 3825–3849. [PDF] [DOI] (Co-First Author, Rank Third)

[J.3] Shuya Hao‡, Maoyin Wang‡, **Leshen Zhang**, Ximeng Lv, Chen Peng, Yuhang Huang, Pei Yuan*, Qing Han*, Gengfeng Zheng*. **Switching Photocatalytic Methane Oxidation Toward Ethanol by Tuning Spin States**.
Angew. Chem. Int. Ed. **2025**, 64, e202510241. [PDF] [DOI] (Third Author)

PREPRINTS & PAPERS UNDER REVIEW

‡EQUAL CONTRIBUTION, *CORRESPONDING AUTHOR

[S.1] **Leshen Zhang**‡, Xujian Wang‡, Taoyu Niu, Xuchao Zhou, Xiao Ma, Guangsheng Liu, Olexandr Isayev*, Junmei Wang*, Wan-Lu Li*. **Incorporating Polarization into ML/MM Endpoint Calculations for Protein–Ligand Affinity Ranking**.
JACS Au, under review. [PDF] [DOI] (Co-First Author, Rank First)

[S.2] Guangsheng Liu‡, **Leshen Zhang**‡, Xiao Ma, Abdulrahman Allangawi, Huabin Zhang*, Wan-Lu Li*. **Local-Environment Populations Govern Size-Dependent OER Activity in Supported Subnanometer IrO_x Clusters**.
Adv. Funct. Mater., under review. [PDF] [DOI] (Co-First Author, Rank Second)

[S.3] Xiao Ma, **Leshen Zhang**, Guangsheng Liu, Abdulrahman Allangawi, Huabin Zhang*, Wan-Lu Li*. **Surface Geometry Governs Cation Effects via Nonlocal Solvent Gating**.
Angew. Chem. Int. Ed., under review. [PDF] [DOI] (Second Author)

[S.4] Guangsheng Liu‡, Xiao Ma‡, **Leshen Zhang**‡, Xujian Wang, Abdulrahman Allangawi, Shaowei Li, Wan-Lu Li*. **Intercalation-Modulated Electronic Regulation of Nitrogen Reduction on Doped MoS_2**.
To be submitted. [PDF] [DOI] (Co-First Author, Rank Third)

[S.5] Guangsheng Liu‡, **Leshen Zhang**‡, Xiao Ma, I-Shan Tsai, Jinyi Zhang, Wan-Lu Li*. **ChemMate: An Autonomous Research Mate for the Full Computational Chemistry Research Lifecycle, from Ideation to Manuscript**.
To be submitted. [PDF] [DOI] (Co-First Author, Rank Second)

[S.6] Tingsong Li‡, Jifeng Wu‡, Abdulrahman Allangawi‡, **Leshen Zhang**, Wan-Lu Li*, Xiangyun Xiao, Miao Hu, Mohd Adnan Khan, Rashed Aleisa, Shouwei Zuo, Heng Guo, Kuo-Wei Huang*, Ying Zhou*, Huabin Zhang*, Magnus Rueping*. **Interfacial Proton Relay Enables CO_2-to-Methanol Electroreduction on Layered Cobalt Phthalocyanine**.
J. Am. Chem. Soc., under review. (Fourth Author)

[S.7] Xuchao Zhou, Xujian Wang, Jiahao Xie, **Leshen Zhang**, Junmei Wang*. **Leveraging Machine Learning Interatomic Potentials for Fast and Accurate AMBER-Compatible Force Field Parameterization**.
J. Chem. Theory Comput., under review. [PDF] [DOI] (Fourth Author)

[S.8] Yunzhi Wang‡, Chengyang Feng‡, Lijie Wang‡, Yang Cheng, **Leshen Zhang**, Yinglu Jia, Lingyun Zhao, Abdulrahman Allangawi, Weiwei Li*, Wan-Lu Li*, Martin Heeney, Magnus Rueping, Omar F. Mohammed*, Huabin Zhang*. **Orchestrated Molecular Intercalation and Surface Pt Exposure for NIR-Driven Hydrogen Evolution**.
Energy Environ. Sci., under review. (Fifth Author)

SELECTED RESEARCH EXPERIENCE

I. First-Principles Calculations for Materials and Catalysis

- **Defect-Mediated Regulation of Interfacial Hydrophobic Transport via Cavitation Thermodynamics**
  [J.1] *PI:* ***Prof. Wan-Lu Li****, UC San Diego, USA*
  Revealed that surface defects regulate interfacial hydrophobic transport by tuning the local cavitation free energy, establishing cavitation thermodynamics as a descriptor linking interfacial water structure to reactivity
- **Site-Resolved C–C Coupling on Restructured Cu Electrodes (ongoing)**
  PI: ***Prof. Anastassia N. Alexandrova****, UC Los Angeles, USA*
  Investigating how kinetically trapped, adsorbate-roughened Cu electrodes catalyze C–C coupling during CO_2 reduction, and using site-resolved KMC simulation to identify the classes of restructured Cu sites that dominate the coupling rate

II. AI-Assisted Scientific Software

- **CatGo: Bridging CLI Coding Agents with Interactive Structure and Workflow Management for Computational Chemistry**
  [J.2] *PI:* ***Prof. Wan-Lu Li****, UC San Diego, USA*
  Established an open-source platform unifying the structure → DFT → HPC → analysis chain, and bridged AI coding agents to its workbench, enabling high-throughput calculations

III. Molecular Modeling: Electrostatics in Enzyme Design

- **Incorporating Polarization into ML/MM Endpoint Calculations for Protein–Ligand Affinity Ranking**
  [S.1] *PI:* ***Prof. Wan-Lu Li****, UC San Diego, and* ***Prof. Junmei Wang****, University of Pittsburgh, USA*
  Developed the PCME method for ML/MM embedding, which feeds conformation-resolved ML charges of the ligand and binding site into the Poisson–Boltzmann solver, accurately predicting binding affinities with high computational efficiency

SKILLS

Theoretical Chemistry Skills

- **First principles:** DFT, GC-DFT, constant-potential simulation, AIMD, enhanced sampling (VASP, CP2K, Gaussian)
- **Molecular simulation:** ML/MM, ML-MD, FEP/TI, MM/PB(GB)SA

Machine Learning Skills

- **Models:** equivariant message-passing graph neural networks, CNNs, RNNs, and symbolic regression (SISSO)
- **Applications:** MLIP training (MACE, UMA, DeePMD), descriptor discovery, and molecular classification

SELECTED HONORS AND AWARDS

- NSFC Grant — Pilot Program for Young Students Basic Research (Undergraduate), National Natural Science Foundation of China 09/2026 – 06/2027
- Fudan University ‘Syensqo’ Scholarship 11/2025
- Fudan University & Chemours ‘Future of Chemistry’ Scholarship 11/2024
- Second Prize, 36th China Chemistry Olympiad (National Final) 11/2022

SELECTED SCIENTIFIC PRESENTATIONS

Oral Presentations

- “Defect-Mediated Regulation of Hydrophobic Transport via Cavitation Thermodynamics” 12/2026 (expected)
  Chinese Chemical Society, The 16th National Conference of Theoretical and Computational Chemistry, Shenzhen, China

PROFESSIONAL SERVICE

Peer Reviewer, *J. Colloid Interface Sci.*, 5 completed reviews 2024 – 2026